

NTT-Friendly Pairing Curve Construction with Readability-Aware Prime Selection over a 1024-bit Field

Tran Duy

University of Engineering and Technology
Vietnam National University, Hanoi
Email: placeholder@vnu.edu.vn

Abstract—Pairing-friendly elliptic curves are fundamental to advanced cryptographic protocols such as BLS signatures and zero-knowledge proof systems. However, the traditional Cocks-Pinch construction produces primes with no guaranteed algebraic structure, limiting their utility for NTT-based proof systems and hindering parameter auditability. We present two contributions: (1) a modified Cocks-Pinch algorithm that constrains the cyclotomic seed to guarantee two-adicity of at least 36 in both the base field and scalar field primes, exceeding the industry-standard BLS12-381 curve; and (2) a multi-criteria readability scoring system that selects human-auditable primes from the valid candidate space without weakening security. We instantiate these techniques to construct a 1024-bit pairing-friendly curve with embedding degree 18, providing 256-bit ECDLP security. A complete Rust implementation demonstrates practical signing operations and validates resistance against MOV, Anomalous, and TNFS attacks.

Index Terms—elliptic curve cryptography, pairing-friendly curves, Cocks-Pinch method, NTT-friendly primes, parameter auditability

Pairing-friendly elliptic curves underpin a growing class of cryptographic protocols, from BLS aggregate signatures [1] to succinct non-interactive zero-knowledge proofs (zk-SNARKs) [2], [3]. These protocols require curves whose scalar field and base field support efficient polynomial arithmetic via the Number Theoretic Transform (NTT), which in turn demands that both field primes satisfy $q \equiv 1 \pmod{2^s}$ for a sufficiently large s .

The Cocks-Pinch method [4] offers unmatched flexibility in constructing pairing-friendly curves with arbitrary embedding degrees. Unlike parametric families such as BN [5] or BLS12 [6], the Cocks-Pinch approach can target specific security levels and field sizes. However, the traditional algorithm treats NTT-friendliness as an afterthought—primes are generated with random two-adicity, typically only 1–3 bits, rendering them unsuitable for modern proof systems.

A separate but related concern is *parameter auditability*. The Dual_EC_DRBG incident [7] demonstrated that opaque cryptographic constants can conceal backdoors. While the Cocks-Pinch method is inherently transparent (all parameters derive from a public seed), the resulting primes are typically inscrutable hexadecimal blobs that resist manual inspection.

Contributions. This paper addresses both concerns:

1. We modify the Cocks-Pinch construction to *guarantee* high two-adicity in both field primes by constraining

the cyclotomic seed T . Specifically, choosing $T \equiv 0 \pmod{2^{12}}$ yields $\nu_2(r-1) \geq 36$ deterministically, while an extended lift search ensures $\nu_2(p-1) \geq 36$. The resulting two-adicity of 36 exceeds BLS12-381’s value of 32.

2. We introduce a *readability-aware prime selection* framework that scores candidate primes on eight structural criteria (binary sparseness, zero limbs, hex density, etc.) and selects the most human-readable valid parameters without sacrificing security.

We instantiate these techniques to produce *Curve1024*: a pairing-friendly curve over a 1024-bit prime field with embedding degree $k = 18$, scalar field order $r \approx 2^{512}$ (256-bit ECDLP security), and NTT support up to length 2^{36} in both fields. A complete Rust implementation validates correctness through 36 tests and demonstrates practical digital signature operations.

Organization. Section 2 reviews background on pairing-friendly curves and the CM method. Section 3 presents our NTT-friendly Cocks-Pinch modification. Section 4 describes the readability scoring system. Section 5 reports implementation results and security analysis. Section 6 discusses related work, and Section 7 concludes.

I. ELLIPTIC CURVES AND PAIRINGS

Let $\mathbb{F}_p$ be a prime field with $p > 3$. An elliptic curve in Short Weierstrass form is $E : y^2 = x^3 + ax + b$ with $4a^3 + 27b^2 \neq 0$. The set $E(\mathbb{F}_p)$ of rational points forms an abelian group under the chord-and-tangent law [8].

A curve is *pairing-friendly* with embedding degree k if there exists a prime-order subgroup of size r such that $r \mid p^k - 1$ but $r \nmid p^i - 1$ for $0 < i < k$. This enables a bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ where $\mathbb{G}_T \subset \mathbb{F}_{p^k}^*$, which is the foundation of BLS signatures and zk-SNARKs.

II. COMPLEX MULTIPLICATION METHOD

The CM method constructs curves with a prescribed group order. Given a CM discriminant $D < 0$, the existence of a curve over $\mathbb{F}_p$ with Frobenius trace t requires the Diophantine equation:

$$4p = t^2 + |D| \cdot y^2$$

to have an integer solution. The j -invariant is obtained from the Hilbert class polynomial $H_D(x)$. For $D = -3$, we have $H_{-3}(x) = x$, giving $j = 0$ and the simple curve form $y^2 = x^3 + b$.

III. TRADITIONAL COCKS-PINCH ALGORITHM

The Cocks-Pinch method [4] constructs pairing-friendly curves by starting from the subgroup order r rather than the field prime p . For embedding degree $k = 18$:

1. **Generate r :** Choose a parameter T and compute $r = \Phi_{18}(T) = T^6 - T^3 + 1$. Verify that r is prime.
2. **Compute square root:** Find $\beta = \sqrt{-3} \pmod{r}$ via Tonelli-Shanks.
3. **Compute CM parameters:** For each i with $\gcd(i, 18) = 1$:

$$t_0 \equiv T^i + 1 \pmod{r}, \quad y_0 \equiv \frac{t_0 - 2}{\beta} \pmod{r}$$

4. **Lift to integers:** Search over $(h_t, h_y) \in [-20, 20]^2$ for:

$$t = t_0 + h_t \cdot r, \quad y = y_0 + h_y \cdot r$$

such that $p = (t^2 + 3y^2)/4$ is a prime of the target size.

The resulting curve has $\rho = \log p / \log r \approx 2$, which is the standard tradeoff of the Cocks-Pinch method compared to parametric families ($\rho = 1$ for BN, $\rho = 1.5$ for BLS12).

IV. TWO-ADICITY AND NTT

The *two-adicity* of a prime q is $\nu_2(q - 1)$, the largest power of 2 dividing $q - 1$. A field $\mathbb{F}_q$ with two-adicity s contains a primitive 2^s -th root of unity, enabling NTT of length up to 2^s . Modern ZK proof systems (Groth16 [2], PLONK [3]) require NTT for polynomial multiplication over both $\mathbb{F}_r$ (constraint evaluation) and $\mathbb{F}_p$ (pairing computation). BLS12-381 achieves two-adicity 32 in both fields [6]. We modify the traditional Cocks-Pinch algorithm to guarantee high two-adicity in both output primes. The key insight is that the algebraic structure of the cyclotomic polynomial Φ_{18} allows deterministic control over the two-adicity of r , while an extended lift search achieves the same for p .

V. NTT-FRIENDLY SCALAR FIELD VIA SEED CONSTRAINT

Since $r = \Phi_{18}(T) = T^6 - T^3 + 1$, we observe:

$$r - 1 = T^6 - T^3 = T^3(T^3 - 1)$$

If $T \equiv 0 \pmod{2^m}$, then $T^3 \equiv 0 \pmod{2^{3m}}$. Since $T^3 - 1$ is odd (as T^3 is even), we obtain:

$$\nu_2(r - 1) = \nu_2(T^3) = 3m$$

Consequence: To guarantee $\nu_2(r - 1) \geq s$, it suffices to choose $T \equiv 0 \pmod{2^{\lceil s/3 \rceil}}$. This is a *deterministic* guarantee requiring no rejection sampling.

For our target of $s = 36$, we set $m = \lceil 36/3 \rceil = 12$, constraining $T \equiv 0 \pmod{2^{12}}$. The search space remains $2^{86-12} = 2^{74}$ candidates—more than sufficient for finding prime r .

VI. NTT-FRIENDLY BASE FIELD VIA EXTENDED LIFT SEARCH

After obtaining a valid r with the desired two-adicity, the base field prime is:

$$p = \frac{(t_0 + h_t r)^2 + 3(y_0 + h_y r)^2}{4}$$

The two-adicity of $p - 1$ depends on the lift parameters (h_t, h_y) . To ensure $\nu_2(p - 1) \geq s_p$, we extend the search grid beyond the default $[-20, 20]^2$:

$$\text{half_range} = 20 + 2^{\max(0, s_p - 3m)}$$

This extension guarantees sufficient coverage of bit patterns in $p - 1$ at positions above $3m$. For $s_p = 36$ and $m = 12$ (so $3m = 36$), the grid remains at $[-21, 21]^2$ —only marginally larger than the default.

VII. ALGORITHM

The complete NTT-friendly Cocks-Pinch construction proceeds as follows:

Input: Embedding degree $k = 18$, target sizes $(|r|, |p|)$, minimum two-adicity s

Output: Primes (p, r) with $\nu_2(r - 1) \geq s$ and $\nu_2(p - 1) \geq s$

1. Set $m \leftarrow \lceil s/3 \rceil$ and step $\leftarrow 2^m$
2. Repeat:
 - a. Choose T as a random multiple of step in the target range
 - b. Compute $r \leftarrow T^6 - T^3 + 1$
 - c. If r is not prime or $|r| \neq \text{target}$, continue
 - d. Compute $\beta \leftarrow \sqrt{-3} \pmod{r}$ (Tonelli-Shanks)
 - e. For each i with $\gcd(i, 18) = 1$:
 - $t_0 \leftarrow T^i + 1 \pmod{r}$
 - $y_0 \leftarrow (t_0 - 2) \cdot \beta^{-1} \pmod{r}$
 - $\text{range} \leftarrow 20 + 2^{\max(0, s - 3m)}$
 - For $(h_t, h_y) \in [-\text{range}, \text{range}]^2$:
 - $t \leftarrow t_0 + h_t \cdot r; y \leftarrow y_0 + h_y \cdot r$
 - $p \leftarrow (t^2 + 3y^2)/4$
 - If $|p| = \text{target}$ AND $\nu_2(p - 1) \geq s$ AND p is prime: return (p, r, t, y)
3. Until max attempts reached

VIII. COMPLEXITY ANALYSIS

The following table compares the traditional and improved algorithms. The NTT constraint increases the average number of attempts by approximately 10–60 $\times$ (from ~ 100 –500 to $\sim 3,000$ –6,000), but the absolute generation time remains under one minute—a one-time cost amortized over the lifetime of the curve parameters.

TABLE I
TRADITIONAL VS. IMPROVED COCKS-PINCH

Metric	Traditional	Improved
Constraint on T	None	$T \equiv 0 \pmod{2^{12}}$
$\nu_2(r-1)$	Random (1–3)	≥ 36 (guaranteed)
$\nu_2(p-1)$	Random (1–3)	≥ 36 (with ext. grid)
Grid size	$[-20, 20]^2$	$[-21, 21]^2$
Avg. attempts	100–500	3,000–6,000
Generation time	2–10 s	30–60 s
NTT support	No	Up to 2^{36}

IX. COMPARISON WITH BLS12-381

BLS12-381 [6], the current industry standard for ZK proof systems, achieves two-adicity 32 in both fields. Our construction achieves 36, enabling NTT of length $2^{36} \approx 6.8 \times 10^{10}$ —a $16\times$ improvement in maximum polynomial degree. The tradeoff is a larger ρ -value (2.0 vs. 1.5) and correspondingly larger field elements, which is acceptable for applications prioritizing long-term security (256-bit ECDLP vs. 128-bit for BLS12-381).

X. MOTIVATION

Cryptographic primes are typically opaque hexadecimal values that resist manual inspection. While the Cocks-Pinch method is inherently transparent—all parameters derive from a public seed T —the resulting primes still appear as random bit strings. This opacity complicates auditing: reviewers cannot easily verify structural properties or detect anomalies by inspection.

The Dual_EC_DRBG incident [7] demonstrated that even standardized constants can harbor backdoors when their generation process is opaque. We propose a complementary approach: among all cryptographically valid candidates, select the one that is most amenable to human inspection.

XI. SECURITY-FIRST DESIGN

Our readability scoring is a *post-filter* applied only to candidates that have already passed all security requirements:

1. Primality (Miller-Rabin with sufficient rounds)
2. Correct bit length (512 for r , 1024 for p)
3. Minimum two-adicity ($\nu_2(q-1) \geq 36$)
4. Valid CM equation ($4p = t^2 + 3y^2$)
5. Embedding degree verification ($r \mid \Phi_{18}(p)$)

The scoring system cannot weaken security because it only selects among pre-validated candidates.

XII. SCORING CRITERIA

Each candidate prime is evaluated on eight independent structural criteria:

The combined score weights the base field prime p at $1.5\times$ the scalar field prime r , since p appears more frequently in implementations (field arithmetic, point encoding, serialization).

TABLE II
READABILITY SCORING CRITERIA

Criterion	Rewards	Score Formula
Binary sparseness	Low Hamming weight	$(n - \text{hw}) \times 2$
Two-adicity	Higher $\nu_2(q-1)$	$\nu_2 \times 3$
Zero limbs	64-bit blocks = 0	$\text{count} \times 60$
Full limbs	Limbs = $0 \times \text{FF} \dots \text{F}$	$\text{count} \times 50$
Longest zero run	Contiguous zero limbs	$\text{run} \times 40$
Hex zero density	Hex digits = 0	$(z - 16) \times 2$
Simple top limb	Few bits in MSB limb	$(32 - \text{hw}) \times 3$
Repeating limbs	Adjacent equal limbs	$\text{pairs} \times 25$

TABLE III
READABILITY METRICS OF SELECTED PARAMETERS

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024	235/512
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16
Hex zero digits	33/256	16/128

XIII. SEARCH PROCEDURE

The readability-aware search operates in two modes:

- **Fast mode:** Returns the first valid (p, r) pair (original behavior).
- **Readable mode:** For each valid r , explores the *full* lift range to find the highest-scoring p . Tracks the global best across N successful attempts (default $N = 50$) within a time budget.

In readable mode, the generator exhaustively searches all (h_t, h_y) combinations for each valid r , scoring every prime candidate that passes the security filter. The best-scoring (p, r) pair across all attempts is retained.

XIV. RESULTS

The following table shows the readability metrics of our selected parameters. Both primes share the tail pattern `...d970_00000001`, a consequence of $T \equiv 0 \pmod{2^{12}}$ propagating through the construction. This shared structure aids visual verification.

The scalar field prime r achieves 8 zero limbs out of 16 (50%), meaning half of its 64-bit representation is zero—a property that simplifies implementation and testing. The shared tail `d970_00000001` = $d_{970} \cdot 2^{36} + 1$ visually confirms the NTT-friendly structure.

XV. IMPLEMENTATION OVERVIEW

We implement the complete system in Rust without external cryptographic dependencies. The implementation comprises:

- A custom `U1024` big integer type with Montgomery multiplication [9]
- Prime field arithmetic (`PrimeFieldElement`) with constant-time Fermat inversion
- Elliptic curve group operations in affine coordinates
- Schnorr and ECDSA signature schemes

TABLE IV
PERFORMANCE BENCHMARKS (NO AVX-512)

Operation	Min	Mean	Max
Key generation	1.629 s	1.687 s	1.757 s
Schnorr sign	1.473 s	1.551 s	1.641 s
Schnorr verify	2.656 s	2.831 s	3.025 s
ECDSA sign	1.524 s	1.606 s	1.711 s
ECDSA verify	2.847 s	3.140 s	3.459 s
Scalar mult. $[k]G$	1.061 s	~1.2 s	1.489 s

TABLE V
SECURITY EVALUATION SUMMARY

Attack	Complexity	Status
Pollard- ρ (ECDLP)	$O(2^{256})$	Secure
MOV reduction	DLP in $\mathbb{F}_{p^{18}}$ (18432-bit)	Secure
Anomalous (SSSA)	Requires $\#E = p$	Immune
TNFS	≥ 250 -bit in $\mathbb{F}_{p^{18}}$	Secure
Invalid curve	Point validation in constructor	Mitigated

- The parameter generation tool (Algorithm 1 + readability scoring)

The test suite contains 36 tests covering arithmetic correctness, group law verification, and attack simulation (MOV, Anomalous, Invalid Curve).

XVI. GENERATED PARAMETERS

The construction produces the following curve (Curve1024):

- Curve equation: $E : y^2 = x^3 + 41$
- Base field: $p \approx 2^{1024}$, $\nu_2(p-1) = 36$
- Scalar field: $r \approx 2^{512}$, $\nu_2(r-1) = 36$
- Embedding degree: $k = 18$
- CM discriminant: $D = -3$
- $\rho = \log p / \log r = 2.0$

XVII. PERFORMANCE BENCHMARKS

The following table reports timing measurements using the Criterion framework (release build, LLVM O3 optimization, 10 samples per measurement).

The performance reflects the 1024-bit field size: each scalar multiplication requires ~512 double-and-add iterations, each involving a Fermat inversion (~1536 Montgomery multiplications). This is approximately 8000 $\times$ slower than optimized secp256k1 implementations, which is expected given the 4 $\times$ larger field and $O(n^2)$ multiplication cost. The system targets offline signing scenarios (legal documents, software releases, PKI certificates) where seconds of latency are acceptable.

XVIII. SECURITY ANALYSIS

The 256-bit ECDLP security exceeds NIST SP 800-57 recommendations through 2030+ [10]. The MOV attack maps ECDLP to DLP in $\mathbb{F}_{p^{18}}$, a field of 18,432 bits—far exceeding the 3,072-bit threshold for 128-bit security. The curve is provably non-anomalous since $\#E(\mathbb{F}_p) = h \cdot r$ with $r \approx 2^{512} \neq p \approx 2^{1024}$.

Limitation: The current affine-coordinate implementation uses variable-time double-and-add, which leaks scalar bits

TABLE VI
COMPARISON WITH STANDARD CURVE FAMILIES

Curve	$ p $	ρ	ν_2	Security
BN254	254	1.0	~1	100-bit
BLS12-381	381	1.5	32	128-bit
KSS18 (generic)	varies	1.33	random	varies
Curve1024	1024	2.0	36	256-bit

through timing. A Montgomery ladder (constant-time) is identified as future work.

XIX. CURVE FAMILY COMPARISON

Pairing-friendly curve taxonomy. Freeman, Scott, and Teske [5] provide a comprehensive classification of pairing-friendly curve constructions, including the Cocks-Pinch method, MNT curves, and polynomial families (BN, BLS, KSS). Their work establishes the theoretical framework but does not address NTT-friendliness as a design criterion.

TNFS security updates. Barbulescu and Duquesne [11] demonstrated that the Tower Number Field Sieve reduces the effective security of many pairing-friendly curves, notably downgrading BN256 from 128-bit to approximately 100-bit security. This motivates our choice of a 1024-bit base field with $k = 18$, yielding a pairing target of 18,432 bits.

BLS12-381. Bowe [6] designed BLS12-381 specifically for zk-SNARKs, achieving two-adicity 32 in both fields through careful parameter selection within the BLS12 family. Our work achieves higher two-adicity (36) using the more flexible Cocks-Pinch method, at the cost of a larger ρ -value.

Pasta curves. The Pallas/Vesta cycle [12] achieves NTT-friendliness through a 2-cycle of curves, enabling recursive proof composition. Unlike our approach, Pasta curves are not pairing-friendly and target a different application domain (Halo 2 recursive proofs).

Nothing-up-my-sleeve numbers. The cryptographic community has long advocated for “nothing-up-my-sleeve” constants (e.g., using digits of π) to demonstrate absence of backdoors. Our readability scoring takes a complementary approach: rather than deriving primes from a fixed public constant, we select the most inspectable prime from a large space of equally secure candidates. Both approaches serve the goal of transparent parameter generation.

Cocks-Pinch extensions. Chen, Hong, and Wang [13] extend the Cocks-Pinch method to embedding degrees 5–8 with optimal Ate pairing computation. Our work focuses on $k = 18$ with the novel addition of NTT constraints and readability scoring. We presented two enhancements to the Cocks-Pinch pairing-friendly curve construction. First, by constraining the cyclotomic seed $T \equiv 0 \pmod{2^{12}}$, we guarantee two-adicity ≥ 36 in the scalar field deterministically, while an extended lift search achieves the same for the base field. This exceeds the industry-standard BLS12-381 (two-adicity 32) at a modest increase in generation time (30–60 s vs. 2–10 s). Second, our readability-aware prime selection scores candidates on eight

structural criteria, producing primes that are more amenable to human inspection without any security degradation.

The resulting Curve1024 provides 256-bit ECDLP security with NTT support up to length 2^{36} in both fields, validated by a complete Rust implementation with 36 passing tests. The system is suitable for offline signing applications requiring long-term security guarantees.

Future work includes: (1) projective/Jacobian coordinates to eliminate per-operation inversions, (2) constant-time Montgomery ladder for side-channel resistance, (3) full optimal Ate pairing implementation over $\mathbb{F}_{p^{18}}$ to enable BLS aggregate signatures, and (4) extension to other embedding degrees ($k = 12, 24$).

XX. AVAILABILITY

The implementation is available as open-source software under the MIT license [14].

REFERENCES

- [1] D. Boneh, B. Lynn, and H. Shacham, “Short signatures from the weil pairing,” in *Advances in Cryptology – ASIACRYPT 2001*. Springer, 2001, pp. 514–532.
- [2] J. Groth, “On the size of pairing-based non-interactive arguments,” in *Advances in Cryptology – EUROCRYPT 2016*, ser. LNCS, vol. 9666. Springer, 2016, pp. 305–326.
- [3] A. Gabizon, Z. J. Williamson, and O. Ciobotaru, “PLONK: Permutations over lagrange-bases for oecumenical noninteractive arguments of knowledge,” Cryptology ePrint Archive, Report 2019/953, 2019.
- [4] C. Cocks and R. G. E. Pinch, “Identity-based cryptosystems based on the weil pairing,” 2001, unpublished manuscript.
- [5] D. Freeman, M. Scott, and E. Teske, “A taxonomy of pairing-friendly elliptic curves,” *Journal of Cryptology*, vol. 23, no. 2, pp. 224–280, 2010.
- [6] S. Bowe, “BLS12-381: New zk-SNARK elliptic curve construction,” <https://electriccoin.co/blog/new-snark-curve/>, 2017.
- [7] D. Shumow and N. Ferguson, “On the possibility of a back door in the NIST SP800-90 Dual Ec Prng,” in *CRYPTO 2007 Rump Session*, 2007.
- [8] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed. Springer, 2009.
- [9] P. L. Montgomery, “Modular multiplication without trial division,” *Mathematics of Computation*, vol. 44, no. 170, pp. 519–521, 1985.
- [10] E. Barker, “Recommendation for key management, part 1: General,” National Institute of Standards and Technology, Tech. Rep. NIST Special Publication 800-57 Part 1 Revision 5, 2020.
- [11] R. Barbulescu and S. Duquesne, “Updating key size estimations for pairings,” *Journal of Cryptology*, vol. 32, pp. 1298–1336, 2019.
- [12] S. Bowe, J. Grigg, and D. Hopwood, “The Pasta curves for Halo 2 and beyond,” <https://electriccoin.co/blog/the-pasta-curves-for-halo-2-and-beyond/>, 2020.
- [13] S. Chen, S. Hong, and K. Wang, “Cocks–pinch curves of embedding degrees five to eight and optimal ate pairing computation,” *Science China Mathematics*, vol. 63, pp. 1519–1538, 2020.
- [14] T. Duy, “curve1024: A 1024-bit elliptic curve cryptography library in Rust,” <https://github.com/Tranduy1dol/curve1024>, 2026, open source, MIT License.